

Seifert fibered 3-manifolds and Turaev-Viro invariants volume conjecture

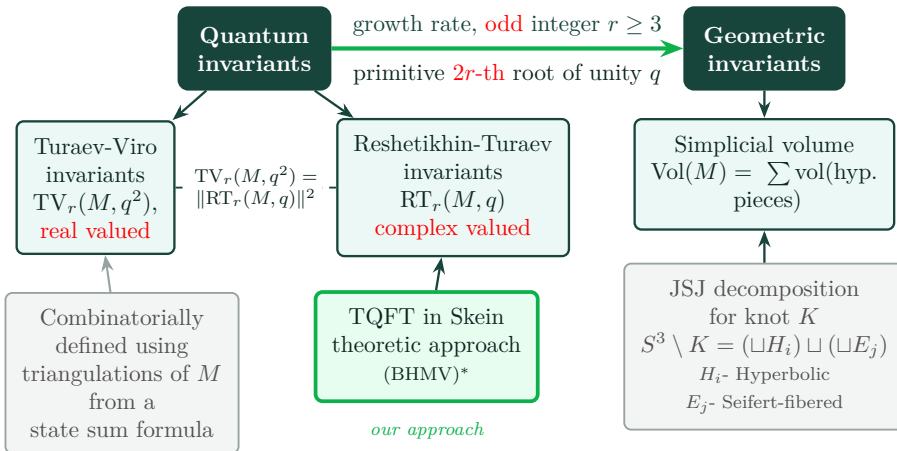
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Joint in parts with Detcherry, Kalfagianni and K. H. Wong

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Volume Conjecture

- 3-manifold M - compact, orientable, $\partial M = \emptyset$ or with toroidal boundary



*BHMV - C. Blanchet, N. Habegger, G. Masbaum and P. Vogel, 1995

Turaev-Viro invariants volume conjecture

Conjecture (Generalization of Q. Chen - T. Yang, 2015)

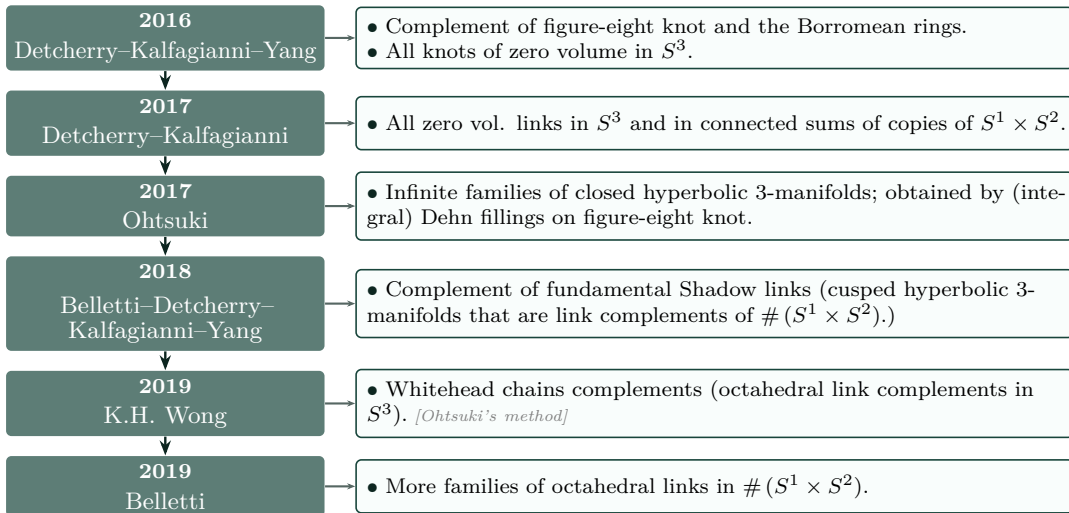
For every compact orientable 3-manifold M , with empty or toroidal boundary, we have

$$LTV(M) := \limsup_{r \rightarrow \infty, r \text{ odd}} \frac{2\pi}{r} \log |TV_r(M, e^{\frac{2\pi i}{r}})| = \text{Vol}(M),$$

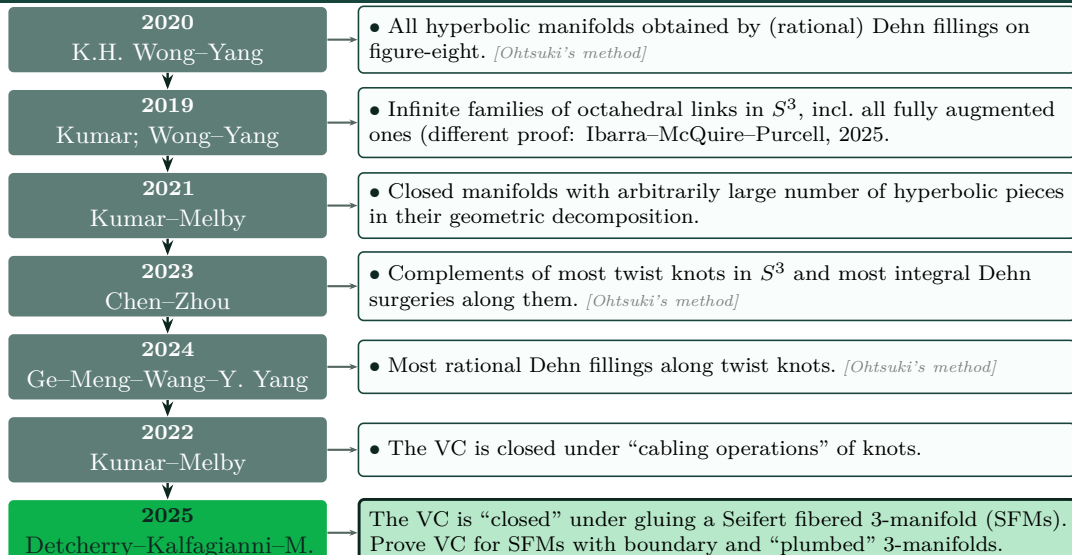
where r runs over all odd integers.

- This is the **3-manifold analogue** of the Kashaev-Murakami-Murakami conjecture (for hyperbolic links in S^3);
- **Hyperbolic M** : Chen–Yang state $LTV(M)$ as a **genuine lim**, and this form is verified for large families of cusped 3-manifolds.
- **General M** : the **lim sup is necessary**. For Seifert fibered 3-manifolds, $TV_r(M; q)$ can vanish for infinitely many r . Some examples (Marasinghe, 2025).
- **Choice q is important!** Eg. for $e^{\frac{\pi i}{r}}$, $LTV(M, e^{\frac{\pi i}{r}}) = 0$, for all 3-manifolds!

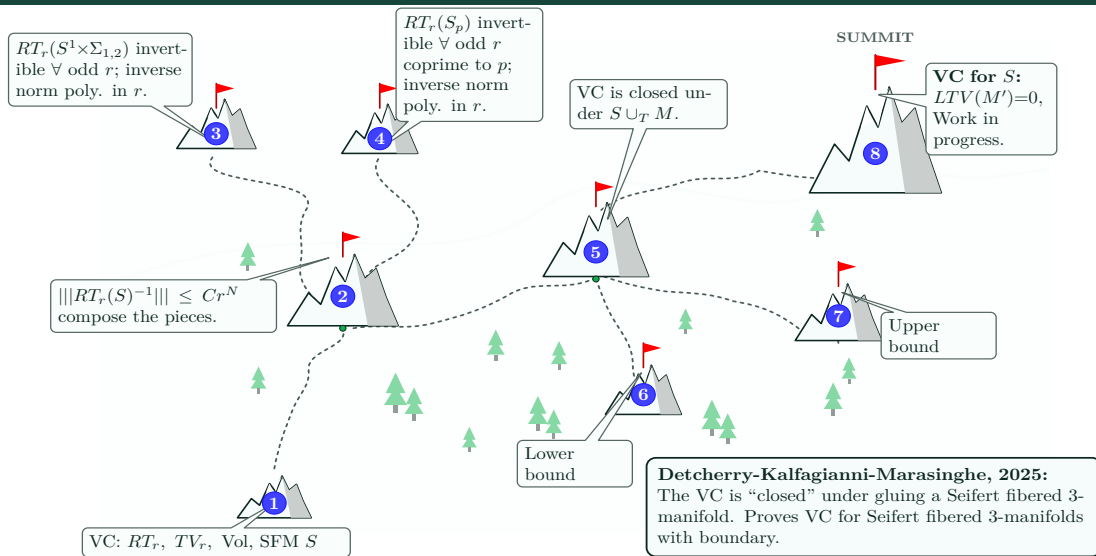
Progress so far



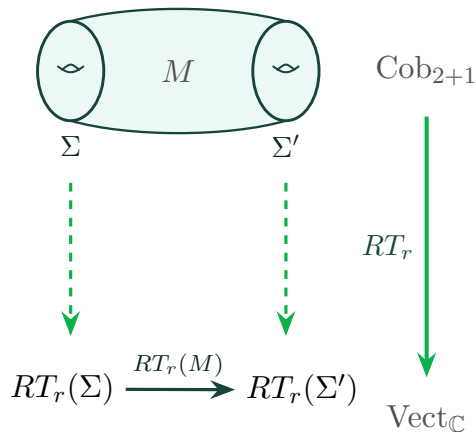
Progress so far: Cont'



The climb to the summit



RT_r invariants (Blanchet, Habegger, Masbaum, Vogel)



Category of (2+1)-dim cobordisms

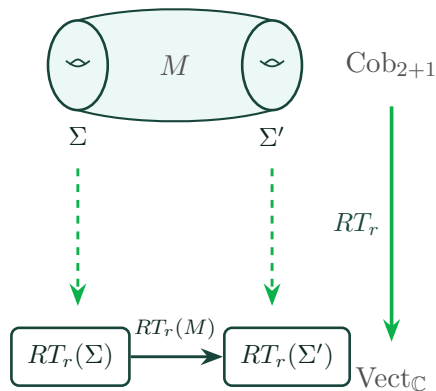
- Objects: oriented closed 2-manifolds
- Morphisms: 3-dim cobordism (M, Σ, Σ') with $\partial M = -\Sigma \sqcup \Sigma'$
- Compositions: gluing (M_1, Σ_1, Σ) and (M_2, Σ, Σ_2) along Σ

Category of finite dim. Hermitian $\mathbb{C}$ -vector spaces

- $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$, with norm induced by Hermitian pairing $\|x\| = \langle x, x \rangle$.
- For $A : V \rightarrow V$ the norm of the operator is $\|A\| = \max_{\|x\|=1} \|A(x)\|$

$SO(3)$ RT_r invariants (BHMV)

- For an odd integer $r \geq 3$ and a primitive $2r$ -th root of unity q



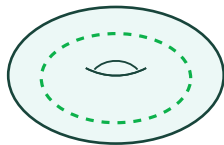
- RT_r takes any **closed oriented surface** Σ to a **finite-dimensional Hermitian $\mathbb{C}$ -vector space** $RT_r(\Sigma)$.
- For a **closed oriented 3-manifold** M , $RT_r(M) \in \mathbb{C}$; if $\partial M \neq \emptyset$, then $RT_r(M) \in RT_r(\partial M)$.
- $RT_r(S^2 \times S^1) = 1$ and $RT_r(S^3) = \frac{2\sin(\frac{2\pi}{r})}{\sqrt{r}}$.
- If M is a cobordism $\Sigma \rightarrow \Sigma'$, then

$$RT_r(M) : RT_r(\Sigma) \rightarrow RT_r(\Sigma')$$

is a linear map, and compositions of cobordisms go to compositions of linear maps (up to powers of q).

$SO(3)$ RT_r invariants of torus T^2

- Here, we recall the basis representing the $\mathbb{C}$ -vector space $RT_r(T^2)$ (BHMV).



$$\{0\} \times S^1$$

- Decorate the **core** $\{0\} \times S^1$ of solid torus with the $i - 1$ Jones-Wenzl idempotent to obtain

$$e_i \in RT_r(T^2).$$

- For $r = 2m + 1$ and q a $2r$ -th root of unity, this process and following the symmetry $e_{m-i} = e_{m+1+i}$ we get the orthonormal basis $e_1, \dots, e_m$ for $RT_r(T^2)$.

- For a genus 1 with two holes $\Sigma_{1,2}$, the corbordism operator

$$RT_r(\Sigma_{1,2} \times S^1) : RT_r(T) \rightarrow RT_r(T')$$

BHMV showed that $RT_r(\Sigma_{1,2} \times S^1)$ is symmetric with respect to above orthonormal basis.

TV_r and RT_r (SO(3) version)

- Let M be a 3-manifold with boundary, and $r \geq 3$ be an odd integer. Then,

$$TV_r(M, q^2) = \eta_r^{-\chi(M)} RT_r(D(M), q),$$

where $D(M) = M \cup_{\partial M} \overline{M}$, $\chi(M)$ is the Euler characteristic of M and $\eta_r = \frac{2\sin(\frac{2\pi}{r})}{\sqrt{r}}$.

- Then, the following can be easily derived using the properties of Hermitian form of $D(M)$.

Theorem (Roberts - closed 3- manifolds; Benedetti , Petronio - manifolds with boundary)

Let M be a compact oriented manifold with toroidal boundary, let $r \geq 3$ be an odd integer and let q be a primitive $2r$ -th root of unity. Then,

$$TV_r(M, q^2) = ||RT_r(M, q)||^2$$

where $|| \cdot ||$ is the natural Hermitian norm on $RT_r(\partial M)$.

Seifert fibered 3-manifold

- Decomposes into a collection of disjoint *fibers* (arranged in a specific way).
- Each fiber has a neighborhood $\cong$ a **fibered solid torus**

$$T(a, b) = (D \times [0, 1]) / \sim$$

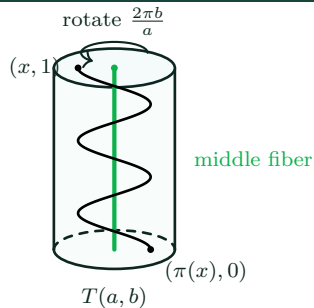
where

- $(x, 1) \sim (\pi(x), 0)$
- π rotates D by $\frac{2\pi b}{a}$, $\gcd(a, b) = 1$
- The image of $\{0\} \times [0, 1]$ in $T(a, b)$ is called the *middle fiber*.

Middle fiber

If $a = 1$: **Ordinary fiber**

If $a > 1$: **Exceptional fiber**



- The 2-orbifold base B is obtained by collapsing each exceptional fiber to a point in $\Sigma \times S^1$, where Σ a surface of genus g .

Standard notation

- We will focus on the orientable orbifold base today.

Oriented Seifert Fibered 3-Manifold - SFM

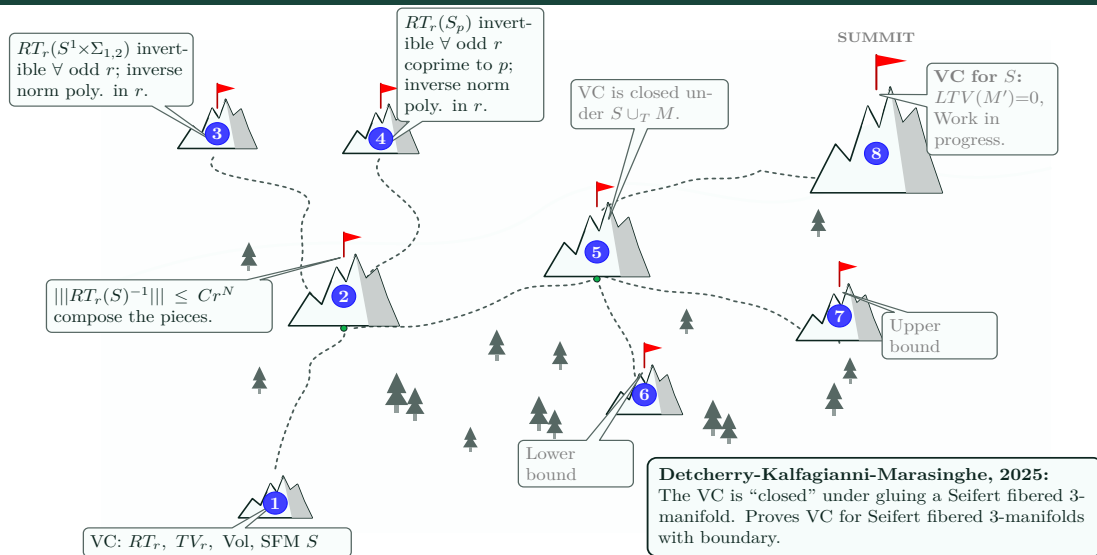
The symbol of an oriented Seifert fibered 3-manifold S that fibers over a 2-orbifold base B is

$$(B; (p_1, q_1), \dots, (p_n, q_n)),$$

where each (p_j, q_j) is a pair of coprime integers with $p_j > 0$, representing an exceptional fiber for $j = 1, \dots, n$ for integer $n > 0$.

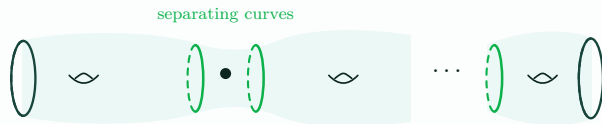
- If $p_j = 1$, for all $j = 1, \dots, n$, then S is an S^1 -bundle over the surface B .
- If the base B has boundary (i.e. B has **holes**), then S also has boundary, which is a **union of tori**.

The climb to the summit



Cutting the base into standard pieces

Base $\Sigma_{g,2}$ (genus g with boundary components)



boundary circles

cut along separating curves



$\Sigma_{1,2}$ (no exceptional fiber)

$\downarrow \times S^1$

Lemma. $RT_r(N = S^1 \times \Sigma_{1,2})$ is invertible for *all* odd $r \geq 3$, and $|||RT_r(N)^{-1}|||$ grows at most polynomially in r . (DKM)



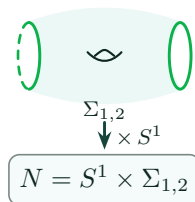
exc. fiber

annulus + 1 exc. fiber

$\downarrow \times S^1$

Lemma. $S_p = (A; \frac{q}{p})$, $p > 1$. Then $RT_r(S_p)$ is invertible for r coprime to p , and $|||RT_r(S_p)^{-1}|||$ grows at most polynomially in r .

- Let $S = (B; \frac{q_1}{p_1}, \dots, \frac{q_n}{p_n})$ with 2 boundary components.
- Goal: $|||RT_r(S)^{-1}|||$ grows at most polynomially in r under certain conditions.
- Cut B along disjoint simple closed separating curves Γ into **annuli with one exc.fiber** or **two-holed tori**.
- Then $\pi^{-1}(\Gamma)$ is a collection of vertical tori wrt fibration in S , under the Seifert fibration $\pi : S \rightarrow B$.
- S_p is a cable space; Kumar–Melby (2023) computed $RT_r(S_p)$ and its norm growth explicitly for r coprime to p .

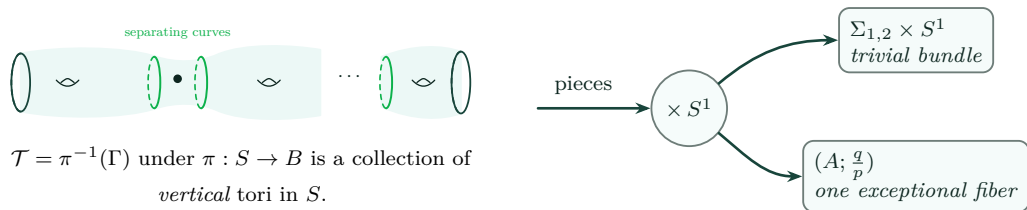
Proof outline: $\Sigma_{1,2}$ case

Lemma (DKM)

Let $N = S^1 \times \Sigma_{1,2}$, the trivial S^1 -bundle over the two-holed torus, with $\partial N = T \cup T'$. Then $RT_r(N)$ is invertible for all odd $r \geq 3$, and the operator norm $|||RT_r(N)^{-1}|||$ grows at most polynomially in all odd integer r .

- $RT_r(N)$ is symmetric wrt the orthonormal basis $e_1, \dots, e_m$ of $RT_r(T^2)$, thus self-adjoint. Then, to bound $|||RT_r(N)^{-1}|||$, it is enough to bound the eigenvalues.
- Also by BHMV, we have the eigenvalues $\lambda_j = \frac{r}{4 \sin^2(\frac{2\pi j}{r})} \neq 0$, $j = 1, \dots, m$.
- Since $\lambda_j \neq 0$, $RT_r(N)$ is invertible for all odd $r \geq 3$.
- Since $|\lambda_j^{-1}| = \frac{4 \sin^2(2\pi j/r)}{r} \leq \frac{4}{r}$, we get $|||RT_r(N)^{-1}||| \leq \frac{4}{r}$ at most polynomial in all odd integer r . □

Combine two pieces



- Each piece is a cobordism $T \rightarrow T'$, so S is a composition of the cobordisms S_i .
- Composition of cobordisms $\rightarrow$ Composition of linear maps (up to powers of q):

$$RT_r(S) = RT_r(S_m) \circ \cdots \circ RT_r(S_0),$$

- Then,

$$RT_r(S)^{-1} = RT_r(S_0)^{-1} \circ \cdots \circ RT_r(S_m)^{-1}.$$

- Each $|||RT_r(S_i)^{-1}|||$ grows at most polynomially in r , and $||| \cdot |||$ is submultiplicative, hence $|||RT_r(S)^{-1}|||$ grows at most polynomially for all odd r coprime to $p_1, \dots, p_n$.

Polynomial growth

Lemma (Detcherry–Kalfagianni–M.)

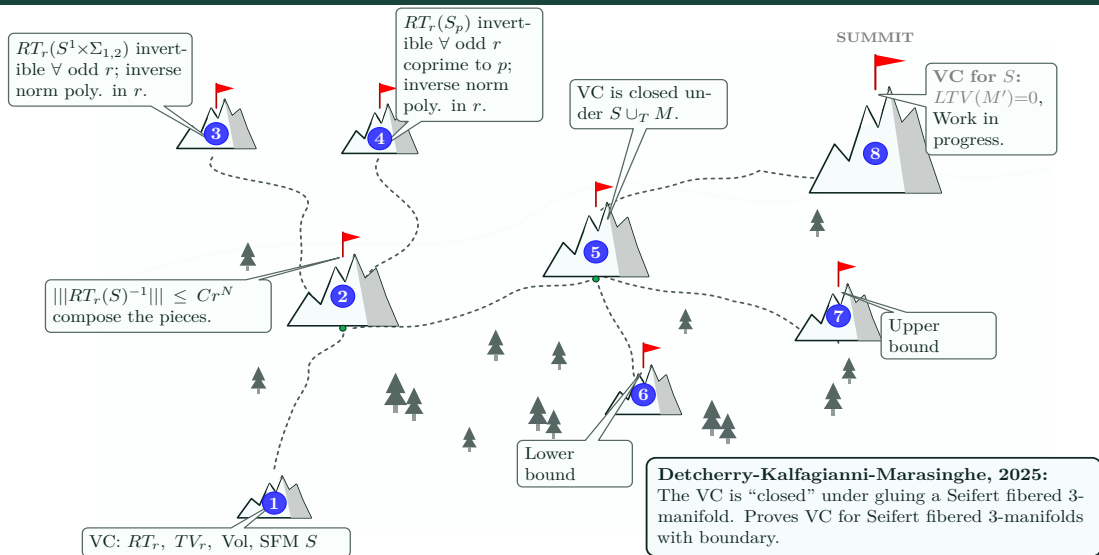
For $\left(S; \frac{q_1}{p_1}, \dots, \frac{q_n}{p_n}\right)$ a Seifert fibered 3-manifold, with $\partial S = T \cup T'$, the linear map

$$RT_r(S) : RT_r(T) \rightarrow RT_r(T),$$

is invertible for all odd r coprime to $p_1, \dots, p_n$. Furthermore, there are constants C and $N > 0$ such that

$$|||RT_r(S)^{-1}||| \leq Cr^N.$$

The climb to the summit



Closed under gluing

Theorem (DKM)

Let $S = \left(B; \frac{q_1}{p_1}, \dots, \frac{q_n}{p_n}\right)$ be a Seifert fibered 3-manifold with *at least two boundary components* and let M be any 3-manifold with toroidal boundary. Then, for any 3-manifold M' obtained by gluing S along a component of ∂S to a component of ∂M , there exist constants A and $K > 0$, such that

$$\frac{r^{-K}}{A} TV_r(M) \leq TV_r(M') \leq Ar^K TV_r(M).$$

Upper inequality holds for all odd r , while lower inequality holds for all r coprime to $p_1, \dots, p_n$.

In particular, if the $\limsup$ in $LTV(M)$ is a limit, then

$$LTV(M') = LTV(M).$$

Lower bound

- We discuss the case when ∂M has one boundary component and ∂S has two boundary components, other cases need more delicate but similar arguments.
- S has two boundary tori T, T' , and

$$RT_r(S) : RT_r(T) \rightarrow RT_r(T')$$

is invertible for all odd r coprime to $p_1, \dots, p_n$, with $|||RT_r(S)^{-1}||| \leq Cr^N \rightarrow (1)$.

- M has a toroidal boundary T^2 , so $RT_r(M) \in RT_r(T^2)$. Gluing S to M gives

$$RT_r(M') = RT_r(S)(RT_r(M)) \implies RT_r(M) = RT_r(S)^{-1}(RT_r(M')).$$

- Apply the operator-norm bound $\|Ax\| \leq |||A||| \|x\|$:

$$\|RT_r(M)\| \leq |||RT_r(S)^{-1}||| \cdot \|RT_r(M')\| \implies |||RT_r(S)^{-1}|||^{-1} \|RT_r(M)\| \leq \|RT_r(M')\|.$$

- By (1), $\frac{r^{-N}}{C} TV_r(M) \leq TV_r(M')$.

Upper bound

Detcherry, Kalfagianni-2017

- ① Let M be a compact orientable 3-manifold that is Seifert fibered. Then $\exists$ constants $A > 0$ and $N > 0$, depending on M , such that

$$TV_r(M, e^{\frac{2\pi i}{r}}) \leq Ar^N \Rightarrow LTV(M) \leq 0.$$

- ② If M' is obtained by gluing any M_1 and M_2 along a boundary torus then:

$$TV_r(M') \leq TV_r(M_1) \cdot TV_r(M_2) \Rightarrow LTV(M') \leq LTV(M_1) + LTV(M_2)$$

- Using 1 and 2 above for $M' = S \cup_T M$,

$$TV_r(M') \leq Ar^K TV_r(M)$$

- Combining both bounds, $\frac{r^{-K}}{A} TV_r(M) \leq TV_r(M') \leq Ar^K TV_r(M)$.
- Finally, this implies that if M satisfies the volume conjecture then M' does as well.

Applications

- Suppose M be a 3-manifold with non-empty boundary such that

$$LTV(M) := \lim_{r \rightarrow \infty, r \text{ odd}} \frac{2\pi}{r} \log |TV_r(M, e^{\frac{2\pi i}{r}})| = \text{Vol}(M),$$

then,

$$LTV(M') := \limsup_{r \rightarrow \infty, r \text{ odd}} \frac{2\pi}{r} \log |TV_r(M', e^{\frac{2\pi i}{r}})| = \text{Vol}(M').$$

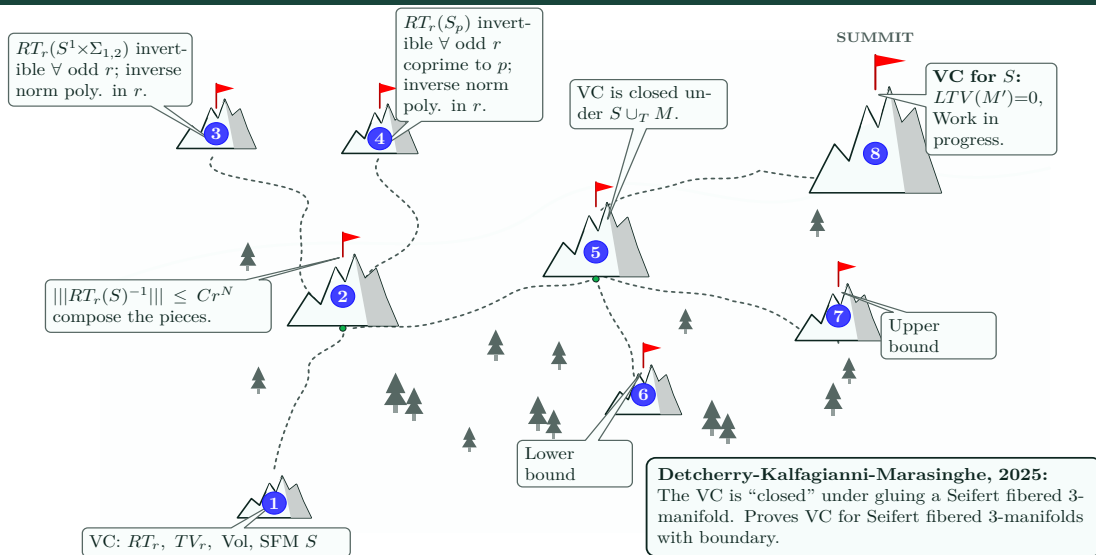
- **Example.** If L is an iterated satellite of the figure-eight with patterns torus links, then

$$LTV(S^3 \setminus L) = \text{Vol}(S^3 \setminus L) \approx 2.0298832.$$

Satellite of a knot K with pattern a torus link = gluing a SFM with at least two boundary components to the boundary torus of the knot complement.

- If $\text{Vol}(S) = 0$, then $LTV(M) = 0 \iff LTV(M') = 0$

The climb to the summit



For SFM S

Corollary

*Suppose that S is an oriented Seifert fibered 3-manifold that either **has a non-empty boundary**, or it is closed and admits an orientation reversing involution. Then we have*

$$LTV(S) := \limsup_{r \rightarrow \infty, r \text{ odd}} \frac{2\pi}{r} \log |TV_r(S, e^{\frac{2\pi i}{r}})| = 0,$$

- By removing a neighborhood of a regular fiber of S with one bound. comp., which is a solid torus $M := D^2 \times S^1$, we get SFM S' that has at least two boundary components.
- Later, one component of $\partial S'$ will be glued to ∂M .
- Also by $TV_r(M, q^2) = \eta_r^{-\chi(M)} RT_r(D(M), q)$, we have

$$TV_r(M) = TV_r(D^2 \times S^1) = RT_r(S^2 \times S^1) = 1,$$

and hence $LTV(M) = \text{Vol}(M) = 0$.

Hyperbolic situation (work in progress)

- Joint work with Ka Ho Wong and Kalfagianni.
- Let $M = (S^3 \setminus K) \cup_{T^2} (S^3 \setminus J)$, $\ell \mapsto \ell + pm$, by integer p positive Dehn twists along the meridian m (here ℓ is the longitude). Then

$$\mathrm{RT}_r(M) = \eta_m^2 \sum_{i=1}^m [i]^2 \left((-1)^{i-1} t^{\frac{i^2-1}{4}} \right)^p \overline{K_i} J_i.$$

Theorem (Marasinghe–Wong–Kalfagianni (in progress))

For M glued from *two figure-eight complements*, an explicit Ohtsuki-type saddle-point analysis gives

$$\mathrm{RT}_r(M) \underset{m \rightarrow \infty}{\sim} \frac{-(2\pi)^{3/2} e^{A_m(1)}}{2\sqrt{-(\pi\sqrt{-1}p - 4\pi\sqrt{3})} 12\pi^2} (m + \tfrac{1}{2})^{1/2} e^{(m+\frac{1}{2})\widetilde{\Phi}(\mathbf{z})},$$

so M satisfies the volume conjecture for every integer p :

$$\lim_{\substack{r \rightarrow \infty \\ r \text{ odd}}} \frac{2\pi}{r} \log |\mathrm{RT}_r(M)|^2 = 2 \mathrm{Vol}(4_1) .$$

Evidence

- Backed by a fast Python implementation evaluating $\text{RT}_r(M)$ (e.g. $m = 1000$ in ~ 0.5 s).
 $4_1 \cup 4_1 \longrightarrow 2 \text{Vol}(4_1) = 4.05976643$
 $4_1 \cup 3_1 \longrightarrow \text{Vol}(4_1) = 2.02988321$

p	Q at $m = 10000$
-3	4.061897561
-2	4.061911944
-1	4.061921240
0	4.061924440
1	4.061921160
2	4.061911784
3	4.061897321

p	Q at $m = 10000$
-3	2.02970964
-2	2.02971117
-1	2.02833364
0	2.02971307
1	2.02830016
2	2.02971248
3	2.02971223



Thank You!
Any Questions?

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